

Multi-Source Scientific Data Request Specification

Wealth, Migration & Retirement Sustainability in Germany (2025-2070)

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Target Completion: Wintersemester 2026/2027 (Archival & Data Destruction: December 2027)

1. Project Architecture & Multi-Source Integration Strategy

1.1 Methodological Positioning

This research platform is designed as a **multi-source statistical integration and calibration framework**, not a direct deterministic person-level linkage of administrative files.

Under German data protection legislation (§ 16 BStatG, § 35 SGB I, § 67 SGB X, and EU GDPR), direct cross-institution micro-linkage across independent federal agencies (DRV, Bundesbank, IAB, Destatis) is legally restricted. Consequently, this study adopts a **calibrated microsimulation architecture**:

- Administrative Histories (FDZ-RV):** Empirically calibrate statutory pension accumulation trajectories (Entgeltpunkte, contribution years, childcare credits).
- Household Balance Sheets (Bundesbank RDSC):** Empirically calibrate household private wealth distributions, housing equity, and liquid buffer dynamics.
- Integration & Wage Trajectories (IAB FDZ):** Empirically estimate employment assimilation curves, language acquisition effects, and credential recognition premiums.
- Macro-Demographic & Population Weights (Destatis FDZ):** Calibrate population weights, household structures, and baseline demographic projections.
- Microsimulation Engine (2025-2070):** Synthesizes these empirical moment distributions into forward-looking individual lifecycle paths to evaluate old-age income adequacy under alternative policy reforms.

1.2 Dual-Track Role of Synthetic Microdata

The pre-existing calibrated synthetic dataset ($N = 50,000$) is preserved as a **development, testing, and open-science reproducibility sandbox**. All preliminary figures derived from synthetic records are explicitly designated as *calibrated baseline simulations*. The real microdata requested from the four FDZs will provide the authoritative empirical parameter estimates for the final thesis.

2. Research Questions Hierarchy

Primary Empirical Question

How do migration timing, labor-market trajectories, and socioeconomic characteristics shape statutory pension accumulation (GRV) and private retirement asset buffers among migrant and native-born populations in Germany?

Secondary Empirical Sub-Questions

- Labor Market & Earnings:** How do years since migration, language proficiency, and qualification recognition relate to gross earnings and annual Entgeltpunkte accumulation?
- Wealth & Portfolio Composition:** How do household net wealth, financial asset portfolios, and homeownership vary across migration cohorts, and what is the velocity of private wealth accumulation over residence duration?

3. **Gender Pension Gap:** How do child-rearing pension credits (§ 56 SGB VI) and part-time employment patterns moderate retirement income disparities between women and men across migrant and native groups?
4. **Safety Net Interaction:** What proportion of individuals in each cohort face autonomous pension outcomes below illustrative subsistence adequacy benchmarks, requiring means-tested basic social assistance (*Grundsicherung im Alter*, SGB XII, Chapter 4)?

Policy Simulation Questions

- How would accelerated qualification recognition (*Fast-Track Anerkennung*) alter lifetime pension points?
- What additional monthly private savings (S^*) are required under alternative capital market return scenarios to achieve a 60% net replacement rate?

3. Operational Cohort Definitions Across Datasets

To ensure statistical rigor, cohort definitions reflect the specific measurement capabilities of each dataset:

Target Cohort	VSKT (FDZ-RV)	IAB-BAMF-SOEP (IAB)	PHF (Bundesbank)	Mikrozensus (Destatis)
1. Native Reference Population	German nationality at birth, domestic contribution history	Non-migrant German reference sample (SOEP Core)	Born in Germany, German citizenship, native parents	German without migration background (<i>ohne Migrationshintergrund</i>)
2. General Migrant (1st Gen)	Foreign nationality / naturalized, delayed contribution start	Immigrants under economic/family categories (M1/M2)	Foreign-born, foreign citizenship or naturalized	First-generation immigrant with direct migration experience
3. General Refugee	Identified via country of origin & registration timing	Humanitarian migration / registered asylum seekers (M3/M4)	Country of origin proxy (high-asylum countries)	Migration reason: humanitarian/asylum
4. Ukrainian Displaced (2022+)	Ukrainian citizenship, first domestic contribution ≥ 2022	Ukrainian refugees registered under § 24 AufenthG (M5)	Arrival year ≥ 2022 , Ukrainian nationality	Ukrainian citizenship, arrival year ≥ 2022
5. Ukrainian Pre-2022 Migrant	Ukrainian citizenship, first domestic contribution < 2022	Ukrainian migrants arriving prior to Feb 2022	Arrival year < 2022 , Ukrainian nationality	Ukrainian citizenship, arrival year < 2022

4. Household vs. Individual Measurement Protocol

- **Household Wealth (PHF):** Wealth is measured and accumulated primarily at the household level. To interface household assets with individual pension entitlements in the microsimulation, household net and financial wealth will be:
 - Analyzed at the household level across household types (single, couple, with/without children).
 - Allocated to individuals using standardized OECD-modified equivalence scales and intra-household sharing rules for scenario testing.
- **Individual Pensions (VSKT):** Statutory pension rights are strictly individualized legal entitlements based on personal contribution records.
- **Separation of Realized Returns from Asset Shares:** Survey data capture static portfolio shares (cash, deposits, equities, real estate). Capital market returns are modeled parametrically via deterministic scenarios and sensitivity bounds.

5. Master Variable Specification

Concept	Required Variable Domain	Primary Source	Secondary Source	Level	Essential?	Purpose
Demographics	Birth year, gender, marital status	VSKT	Mikrozensus / IAB	Person	Yes	Cohort segmentation, life expectancy
Migration	Nationality, arrival year, country of birth	IAB / Mikrozensus	VSKT (citizenship)	Person	Yes	Residence duration calculation
Legal Status	§ 24 AufenthG, asylum status, work permit	IAB-BAMF-SOEP	Mikrozensus	Person	Yes	Refugee cohort isolation
Pension Biography	Contribution months, credited periods, non-contributory times	VSKT	--	Person	Yes	Wartezeit (5, 35, 45 yrs) calculation
Pension Points (EP)	Total EP, EP from earnings, EP from child-rearing (§ 56)	VSKT	--	Person	Yes	Primary pension calculation ($R = EP * ZF * AR$)
Basic Pension	Grundrentenzeiten, Grundrenten-EP (§ 76g)	VSKT	--	Person	Yes	SGB XII § 82a disregard modeling
Labor & Wages	Monthly gross/net wages, full/part-time status	IAB / VSKT	Mikrozensus	Person	Yes	Mincer wage equations, contribution base
Education & Recognition	Foreign degree, formal recognition status, deskilling indicator	IAB-BAMF-SOEP	Mikrozensus	Person	Yes	Qualification wage penalty estimation
Language	CEFR proficiency levels (A1 to C2)	IAB-BAMF-SOEP	--	Person	Yes	Human capital wage covariate
Net Wealth	Total household assets, liabilities, net worth	Bundesbank PHF	EVS (partial)	Household	Yes	Wealth gap estimation, decumulation
Liquid Buffers	Bank deposits, cash, savings accounts	Bundesbank PHF	EVS	Household	Yes	SGB XII Schonvermögen (EUR 10,000) threshold
Equity Assets	Stocks, mutual funds, ETFs	Bundesbank PHF	--	Household	Yes	Portfolio composition & real return modeling
Housing	Homeownership dummy, mortgage balance, property value	Bundesbank PHF	Mikrozensus	Household	Yes	Shelter cost mitigation in retirement
Private Pension	Riester, Rürup, occupational pension contracts	Bundesbank PHF	VSKT (partial)	Household	Yes Useful	Multi-pillar retirement modeling

Concept	Required Variable Domain	Primary Source	Secondary Source	Level	Essential?	Purpose
Household Budget	Housing costs (rent/heating), basic living expenses	Destatis Mikrozensus / EVS	--	Household	Highly Useful	Calibration of illustrative subsistence benchmark
Remittances	Financial transfers abroad (amount, frequency)	IAB-BAMF-SOEP	--	Household	Optional	Savings capacity constraint adjustment

6. Access Modes & Institutional Protocols

1. **FDZ-RV:** Scientific Use File (SUF) of the *Versichertenkontenstichprobe (VSKT)*. Where detailed migration identifiers require higher granularity, remote execution via FDZ-RV standard procedures will be requested.
2. **Bundesbank RDSC:** Scientific Use File (SUF) / Off-site data or Remote Data Execution (RDSC JoSuA/LISS environment) of the *Panel on Household Finances (PHF)*.
3. **IAB FDZ:** Scientific Use File (SUF) of the *IAB-BAMF-SOEP Befragung von Geflüchteten* and *IAB-SOEP Migrationsstichproben*.
4. **Destatis FDZ:** Scientific Use File (SUF) of the *Mikrozensus* (focus: population weighting) and optional *EVS* (focus: consumption/budgets).